Assignment 4

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71 VLSI

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(1) What is the ROC of Z-transform function? With illustrative examples discuss the properties of ROC.

The Region of Convergence or ROC is the range of values of the complex variable z for which the Z transform summation converges.

For a discrete-time signal $x[n]$, the Z transform is defined as

$$X(z) = \sum_{-\infty}^{\infty} x[n] z^{-n}$$

ROC is the set of all z in the complex plane for which the summation converges to a finite value.

Properties:

(1) ROC is a ring or a disk in the z plane centered at origin, can extend outward to ∞ & inward to 0

Eg → For $X(z) = \frac{1}{(1-0.5z^{-1})(1-0.8z^{-1})}$

the poles are at $z = 0.5$ & $z = 0.8$

ROC could be
- $|z| > 0.8$
- $|z| < 0.5$
- $0.5 < |z| < 0.8$

(2) ROC does not include poles

Eg → $X(z) = \frac{1}{1-0.5z^{-1}}$

there's a pole at $z = 0.5$

& ROC is $|z| > 0.5$

(3) For finite duration signals, the ROC is entire z plane, except possibly at $z = 0$ or $z = \infty$

Eg $x[n] = 1, 2, 3$

$X(z) = 1 + 2z^{-1} + 3z^{-2}$ → converges for all except $z = 0$ or ∞

(4) For a right sided (causal) sequence ROC is of the form $|z| > \gamma$

(4) For a right sided sequence ROC is of the form $|z| > \gamma$ (causal)

Eg → $x[n] = (0.5)^n u[n]$

$X(z) = \frac{1}{1 - 0.5z^{-1}}$ for $|z| > 0.5$

(5) For a left sided sequence ROC is of the form $|z| < \gamma$ (anti causal)

Eg → $x[n] = -(0.5)^n u[-n-1]$

$X(z) = \frac{1}{1 - 0.5z^{-1}}$ for $|z| < 0.5$

(6) For a two sided sequence ROC lies between two circles

Eg → $x[n] = (0.5)^n u[n] + (0.8)^n u[-n-1]$

$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 0.8z^{-1}}$

(7) ROC must include unit circle ($|z| = 1$) for stability of $x[n]$

Eg $x[n] = (0.5)^n u[n]$, ROC $= |z| > 0.5$
→ includes unit circle ⇒ Stable

$x[n] = (1.5)^n u[n]$, ROC $= |z| > 1.5$
→ does not include unit circle ⇒ unstable

(2) Explain the time shift property of Z-transform. Highlight the change in ROC due to time shifting. Illustrate with one example for (i) No change in ROC (ii) z = 0 gets excluded (iii) |z| = ∞ gets excluded.

$x[n] \xleftrightarrow{z} X(z)$

$x[n - n_0] \xleftrightarrow{z} z^{-n_0} X(z)$

Effect of time shifting on ROC

→ ROC of $X(z)$ remains same except if

- sequence is delayed
  then ROC may exclude $z = 0$

- sequence is delayed
  then ROC may exclude $z=0$
- sequence is advanced
  then ROC may exclude $z=\infty$

→ For signals that are finite or bounded ROC may remain unchanged.

(i) No change in ROC

$x[n] = \alpha^n u[n] \quad |\alpha|<1$

$X(z) = \frac{1}{1-\alpha z^{-1}}$, ROC: $|z|>|\alpha|$

now delaying

$x[n-2] = \alpha^{n-2} u[n-2]$

$X_1(z) = z^{-2} X(z) = \frac{z^{-2}}{1-\alpha z^{-1}}$, ROC: $|z|>|\alpha|$ unchanged

(ii) $z=0$ gets excluded

$x[n] = \{1,2,3\}$

$n = 0, 1, 2$

then $X(z) = 1 + 2z^{-1} + 3z^{-2}$

ROC: entire plane except $z=0$

now delay $x[n-1]$

$X_1(z) = z^{-1} X(z) = z^{-1} + 2z^{-2} + 3z^{-3}$

ROC: all z except $z=0$

$z=0$ gets excluded

(iii) $|z| = \infty$ gets excluded

$x[n] = \{1,2,3\} \quad n = 0,1,2$

$x[n+1]$

$X_2(z) = zX(z) = z + 2 + 3z^{-1}$

ROC: all z except $z=\infty$

$z=\infty$ gets

$z = \infty$ gets excluded

(6) Given: $x(t) = 4cos(8\pi t) + 6sin(12\pi t) + cos(20\pi t)$

Determine the minimum sampling frequency.

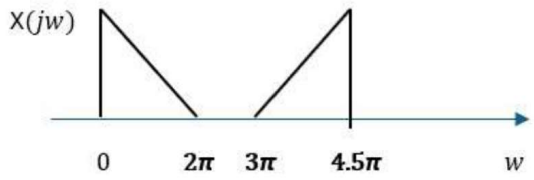
min^m sampling frequency
$= 2 \times$ (max^m freq. component)
(nyquist theorem)

$\cos(8\pi t) \rightarrow \omega_1 = 8\pi, f_1 = 4Hz$ $\sin(12\pi t) \rightarrow \omega_2 = 12\pi$, $f_2 = 6Hz$

$\cos(20\pi t) \rightarrow \omega_3 = 20\pi, f_3 = 10Hz$

max freq component $= 10Hz$
min^m Sampling freq $= 20Hz$

(7)



Find the maximum sampling interval Ts.

max frequency $\omega_m = 4.5\pi$
max^m Sampling interval is given by nyquist theorem

$\omega_s \geq 2\omega_m$

$\omega_s = 2\pi / T_s$

$\frac{2\pi}{T_s} \geq 2 \times 4.5\pi$

$T_s \geq 0.222\ s$

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The spectrum of a signal $X(j\omega)$ is band-limited to $|\omega| \leq \omega_m$.
The signal is sampled with period T_s.

a) Write the expression for the sampled spectrum $X_s(j\omega)$.
b) Determine the condition on T_s to avoid aliasing.
c) Design the ideal reconstruction filter $H_r(j\omega)$ required to perfectly recover $X(j\omega)$

a) $x_s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow X_s(jw) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$

$\omega_s = \frac{2\pi}{T_s}$

b) To avoid aliasing

$\omega_s \geq 2\omega_m$

$\frac{2\pi}{T_s} \geq 2\omega_m$

$\Rightarrow T_s \leq \frac{\pi}{\omega_m}$

c) Low pass filter

① Passes original base band copy without distortion

② Completely blocks all spectral copies centered at nonzero multiples of ω_s

$$H(jw) = \begin{cases} T_s & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

ω_c must satisfy

$\omega_m < \omega_c < \omega_s - \omega_m$